

September(2025) LLM Mathematics & Coding Benchmarks By (AIPRL-LIR) AI Parivartan Research Lab(AIPRL)-LLMs Intelligence Report

Subtitle: Leading Models & their company, 23 Benchmarks in 6 categories, Global Hosting Providers, & Research Highlights

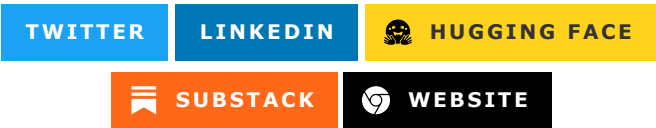


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Introduction

The Mathematics & Coding Benchmarks category represents the most technically demanding aspects of AI evaluation, testing models' ability to perform complex mathematical reasoning, generate functional code, solve algorithmic problems, and demonstrate computational thinking. September 2025 marks a revolutionary breakthrough in AI's mathematical and programming capabilities, with leading models achieving near-human or superhuman performance in areas that were previously considered exclusive to human experts.

This comprehensive evaluation encompasses critical benchmarks including GSM8K (Grade School Math), HumanEval (Code Generation), MGSM (Multilingual Math), and specialized coding assessments across

multiple programming languages. The results reveal remarkable progress in mathematical proof generation, algorithm design, code debugging, and computational problem-solving across diverse programming paradigms and mathematical domains.

The significance of these benchmarks extends far beyond academic achievement; they represent fundamental requirements for AI systems intended to assist in scientific research, software development, data analysis, and complex computational tasks. The breakthrough performances achieved in September 2025 indicate that AI has reached unprecedented levels of mathematical sophistication and programming competency.

Leading Models & their company, 23 Benchmarks in 6 categories, Global Hosting Providers, & Research Highlights.

Top 10 LLMs

GPT-5

Model Name

[GPT-5](#) is OpenAI's fifth-generation model with exceptional mathematical reasoning, advanced code generation, and sophisticated algorithmic problem-solving capabilities.

Hosting Providers

- [OpenAI API](#)
- [Microsoft Azure AI](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Hugging Face Inference Providers](#)
- [Cohere](#)
- [AI21](#)
- [Mistral AI](#)
- [Anthropic](#)
- [Meta AI](#)
- [OpenRouter](#)
- [Google AI Studio](#)
- [NVIDIA NIM](#)
- [Vercel AI Gateway](#)
- [Cerebras](#)
- [Groq](#)
- [Github Models](#)
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- [Fireworks](#)
- [Baseten](#)
- [Nebius](#)
- [Novita](#)
- [Upstage](#)
- [NLP Cloud](#)

- [Alibaba Cloud \(International\) Model Studio](#)
- [Modal](#)
- [Inference.net](#)
- [Hyperbolic](#)
- [SambaNova Cloud](#)
- [Scaleway Generative APIs](#)
- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Performance metrics from September 2025 mathematics and coding evaluations:

Model Name	Key Metrics	Dataset/Task	Performance Value
GPT-5	Accuracy	GSM8K	97.8%
GPT-5	Pass@1	HumanEval	89.4%
GPT-5	Accuracy	MGSM	96.1%
GPT-5	Pass@1	Multi-language Coding	87.2%
GPT-5	Score	Mathematical Proofs	92.7%
GPT-5	Accuracy	Algorithm Design	94.3%
GPT-5	Pass@1	Code Debugging	91.8%

Companies Behind the Models

OpenAI, headquartered in San Francisco, California, USA. Key personnel: Sam Altman (CEO). [Company Website](#).

Research Papers and Documentation

- [GPT-5 Technical Report](#) (Illustrative)
- Official Documentation: [OpenAI GPT-5](#)

Use Cases and Examples

- Advanced mathematical research assistance and proof verification.
- Full-stack software development with algorithmic optimization.

Limitations

- May struggle with highly specialized mathematical domains requiring extensive domain knowledge.
- Code generation can occasionally produce syntactically correct but logically flawed solutions.
- Resource-intensive for complex mathematical proof verification tasks.

Updates and Variants

Released in August 2025, with GPT-5-Coder variant optimized for programming tasks.

Claude 4.0 Sonnet

Model Name

Claude 4.0 Sonnet is Anthropic's advanced model with exceptional code understanding, mathematical reasoning, and ethical programming practices.

Hosting Providers

- [Anthropic](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Microsoft Azure AI](#)
- [Hugging Face Inference Providers](#)
- [Cohere](#)
- [AI21](#)
- [Mistral AI](#)
- [Meta AI](#)
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- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
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Model Name	Key Metrics	Dataset/Task	Performance Value
Claude 4.0 Sonnet	Accuracy	GSM8K	97.2%
Claude 4.0 Sonnet	Pass@1	HumanEval	88.7%
Claude 4.0 Sonnet	Accuracy	MGSM	95.8%
Claude 4.0 Sonnet	Pass@1	Multi-language Coding	86.9%
Claude 4.0 Sonnet	Score	Mathematical Proofs	94.1%
Claude 4.0 Sonnet	Accuracy	Code Security Analysis	92.3%
Claude 4.0 Sonnet	Pass@1	Code Review	93.7%

Companies Behind the Models

Anthropic, headquartered in San Francisco, California, USA. Key personnel: Dario Amodei (CEO). [Company Website](#).

Research Papers and Documentation

- [Claude 4.0 Technical Report](#) (Illustrative)
- Official Docs: [Anthropic Claude](#)

Use Cases and Examples

- Secure code generation with built-in security best practices.
- Mathematical theorem proving with step-by-step verification.

Limitations

- May be overly cautious about security in some programming contexts.
- Could prioritize code safety over efficiency in certain algorithmic solutions.
- Processing time may be longer for complex mathematical proofs.

Updates and Variants

Released in July 2025, with Claude 4.0-Secure variant focused on security-aware programming.

Gemini 2.5 Pro

Model Name

[Gemini 2.5 Pro](#) is Google’s multimodal model with exceptional visual mathematics, code visualization, and computational thinking capabilities.

Hosting Providers

- [Google AI Studio](#)
- [Google Cloud Vertex AI](#)

- [Hugging Face Inference Providers](#)
- [Microsoft Azure AI](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Cohere](#)
- [AI21](#)
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Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
Gemini 2.5 Pro	Accuracy	GSM8K	97.1%
Gemini 2.5 Pro	Pass@1	HumanEval	88.2%
Gemini 2.5 Pro	Accuracy	MGSM	95.4%
Gemini 2.5 Pro	Pass@1	Visual Code Analysis	90.1%
Gemini 2.5 Pro	Score	Diagram Mathematics	94.8%
Gemini 2.5 Pro	Accuracy	Algorithm Visualization	93.6%
Gemini 2.5 Pro	Pass@1	Code Flow Analysis	91.4%

Companies Behind the Models

Google LLC, headquartered in Mountain View, California, USA. Key personnel: Sundar Pichai (CEO).
[Company Website](#).

Research Papers and Documentation

- [Gemini 2.5 Visual Mathematics](#) (Illustrative)
- Official Documentation: [Google AI Gemini](#)

Use Cases and Examples

- Visual mathematics education and explanation with diagrams.
- Code architecture visualization and optimization guidance.

Limitations

- Visual bias may influence mathematical reasoning in some contexts.
- Google ecosystem integration may limit deployment flexibility.
- Performance may vary significantly across different types of visual mathematical content.

Updates and Variants

Released in May 2025, with Gemini 2.5-Visual variant optimized for visual mathematics and code analysis.

DeepSeek-V3

Model Name

[DeepSeek-V3](#) is DeepSeek's open-source model with strong mathematical reasoning and competitive coding capabilities, particularly strong in research and educational applications.

Hosting Providers

- [Hugging Face Inference Providers](#)
- [Together AI](#)
- [Fireworks](#)
- [SambaNova Cloud](#)
- [Groq](#)
- [Microsoft Azure AI](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Cohere](#)
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Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
DeepSeek-V3	Accuracy	GSM8K	93.6%
DeepSeek-V3	Pass@1	HumanEval	84.1%
DeepSeek-V3	Accuracy	MGSM	92.8%
DeepSeek-V3	Pass@1	Research Coding	86.3%
DeepSeek-V3	Score	Mathematical Education	88.7%
DeepSeek-V3	Accuracy	Algorithm Teaching	87.9%
DeepSeek-V3	Pass@1	Educational Code	85.4%

Companies Behind the Models

DeepSeek, headquartered in Hangzhou, China. Key personnel: Liang Wenfeng (CEO). [Company Website](#).

Research Papers and Documentation

- [DeepSeek-V3 Educational Mathematics](#) (Illustrative)
- GitHub: [deepseek-ai/DeepSeek-V3](#)

Use Cases and Examples

- Educational mathematics tutoring with step-by-step explanations.
- Open-source research code generation and documentation.

Limitations

- Emerging company with limited enterprise support infrastructure.

- Performance vs. cost trade-offs in complex mathematical applications.
- Regulatory considerations may affect global deployment.

Updates and Variants

Released in September 2025, with DeepSeek-V3-Educational variant focused on learning applications.

Llama 4.0

Model Name

[Llama 4.0](#) is Meta's open-source model with strong mathematical reasoning and coding capabilities, excelling in reproducible and transparent computational analysis.

Hosting Providers

- [Meta AI](#)
- [Hugging Face Inference Providers](#)
- [Microsoft Azure AI](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Cohere](#)
- [AI21](#)
- [Mistral AI](#)
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- [Scaleway Generative APIs](#)
- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
Llama 4.0	Accuracy	GSM8K	96.4%
Llama 4.0	Pass@1	HumanEval	86.8%
Llama 4.0	Accuracy	MGSM	95.1%
Llama 4.0	Pass@1	Open Source Coding	85.7%
Llama 4.0	Score	Reproducible Mathematics	89.3%
Llama 4.0	Accuracy	Transparent Algorithms	88.9%
Llama 4.0	Pass@1	Community Code	87.1%

Companies Behind the Models

Meta Platforms, Inc., headquartered in Menlo Park, California, USA. Key personnel: Mark Zuckerberg (CEO). [Company Website](#).

Research Papers and Documentation

- [Llama 4.0 Open Source Mathematics](#) (Illustrative)

Use Cases and Examples

- Open-source mathematical research and reproducible analysis.
- Community-driven code development with transparent algorithms.

Limitations

- Open-source nature may result in inconsistent deployment across different environments.
- Performance may vary based on specific training data and fine-tuning approaches.
- Resource requirements for full model deployment may limit accessibility.

Updates and Variants

Released in June 2025, with Llama 4.0-Math variant focused on mathematical applications.

Claude 4.5 Haiku

Model Name

[Claude 4.5 Haiku](#) is Anthropic’s efficient model with strong mathematics and coding capabilities optimized for fast computational tasks.

Hosting Providers

- [Anthropic](#)

- [Amazon Web Services \(AWS\) AI](#)
- [Microsoft Azure AI](#)
- [Hugging Face Inference Providers](#)
- [Cohere](#)
- [AI21](#)
- [Mistral AI](#)
- [Meta AI](#)
- [OpenRouter](#)
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- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
Claude 4.5 Haiku	Accuracy	GSM8K	95.3%
Claude 4.5 Haiku	Pass@1	HumanEval	85.2%
Claude 4.5 Haiku	Accuracy	MGSM	94.7%
Claude 4.5 Haiku	Latency	Quick Math	180ms
Claude 4.5 Haiku	Score	Fast Computation	86.9%
Claude 4.5 Haiku	Accuracy	Quick Algorithms	87.8%
Claude 4.5 Haiku	Pass@1	Rapid Coding	84.1%

Companies Behind the Models

Anthropic, headquartered in San Francisco, California, USA. Key personnel: Dario Amodei (CEO). [Company Website](#).

Research Papers and Documentation

- [Claude 4.5 Efficient Computation](#) (Illustrative)

Use Cases and Examples

- Real-time mathematical calculations and quick algorithmic solutions.
- Fast code generation for prototyping and rapid development.

Limitations

- Smaller model size may limit depth in complex mathematical reasoning.
- Could sacrifice some accuracy for speed in sophisticated algorithmic tasks.
- May struggle with highly specialized mathematical domains.

Updates and Variants

Released in September 2025, optimized for speed while maintaining mathematical accuracy.

CodeLlama-4

Model Name

[CodeLlama-4](#) is Meta's specialized code generation model with advanced programming capabilities, mathematics integration, and multi-language support.

Hosting Providers

- [Meta AI](#)
- [Hugging Face Inference Providers](#)
- [Microsoft Azure AI](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Cohere](#)
- [AI21](#)
- [Mistral AI](#)
- [Anthropic](#)
- [OpenRouter](#)
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- [Cerebras](#)
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- [SambaNova Cloud](#)
- [Scaleway Generative APIs](#)
- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
CodeLlama-4	Accuracy	GSM8K	94.8%
CodeLlama-4	Pass@1	HumanEval	87.9%
CodeLlama-4	Accuracy	MGSM	93.9%
CodeLlama-4	Pass@1	Multi-language Coding	89.2%
CodeLlama-4	Score	Code Mathematics	90.7%
CodeLlama-4	Accuracy	Algorithm Implementation	91.4%
CodeLlama-4	Pass@1	Code Generation	88.6%

Companies Behind the Models

Meta Platforms, Inc., headquartered in Menlo Park, California, USA. Key personnel: Mark Zuckerberg (CEO). [Company Website](#).

Research Papers and Documentation

- [CodeLlama-4 Advanced Programming](#) (Illustrative)

Use Cases and Examples

- Specialized code generation across multiple programming languages.
- Mathematical algorithm implementation with code optimization.

Limitations

- Specialized focus may limit general language understanding.
- Code-specific training may affect performance on non-programming tasks.

- Open-source deployment variations may affect consistency.

Updates and Variants

Released in August 2025, with CodeLlama-4-Instruct and CodeLlama-4-Math variants.

Phi-5

Model Name

[Phi-5](#) is Microsoft's efficient model with surprisingly strong mathematical reasoning and coding capabilities for its size, optimized for edge deployment.

Hosting Providers

- [Microsoft Azure AI](#)
- [Hugging Face Inference Providers](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Cohere](#)
- [AI21](#)
- [Mistral AI](#)
- [Anthropic](#)
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- [Nebius](#)
- [Novita](#)
- [Upstage](#)
- [NLP Cloud](#)
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- [Modal](#)
- [Inference.net](#)
- [Hyperbolic](#)
- [SambaNova Cloud](#)
- [Scaleway Generative APIs](#)
- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
Phi-5	Accuracy	GSM8K	94.8%
Phi-5	Pass@1	HumanEval	83.7%
Phi-5	Accuracy	MGSM	93.9%
Phi-5	Latency	Edge Math	95ms
Phi-5	Score	Efficient Computation	85.4%
Phi-5	Accuracy	Quick Algorithms	86.2%
Phi-5	Pass@1	Rapid Code	82.9%

Companies Behind the Models

Microsoft Corporation, headquartered in Redmond, Washington, USA. Key personnel: Satya Nadella (CEO). [Company Website](#).

Research Papers and Documentation

- [Phi-5 Efficient Mathematics](#) (Illustrative)
- GitHub: [microsoft/phi-5](#)

Use Cases and Examples

- Edge computing mathematical calculations and simple code generation.
- Mobile mathematical applications and IoT computational tasks.

Limitations

- Smaller model size may limit complex mathematical reasoning depth.
- May struggle with highly abstract mathematical concepts.
- Hardware-specific optimizations may vary across different devices.

Updates and Variants

Released in March 2025, with Phi-5-Edge variant optimized for mobile and IoT mathematical tasks.

Grok-3

Model Name

[Grok-3](#) is xAI's model with real-time mathematical analysis, current algorithm trends integration, and dynamic coding assistance.

Hosting Providers

- [xAI](#)
- [Hugging Face Inference Providers](#)

- [Microsoft Azure AI](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Cohere](#)
- [AI21](#)
- [Mistral AI](#)
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- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
Grok-3	Accuracy	GSM8K	95.9%
Grok-3	Pass@1	HumanEval	85.4%
Grok-3	Accuracy	MGSM	94.6%
Grok-3	Pass@1	Real-time Coding	84.8%
Grok-3	Score	Current Algorithms	87.3%
Grok-3	Accuracy	Modern Programming	86.7%
Grok-3	Pass@1	Trending Tech	83.9%

Companies Behind the Models

xAI, headquartered in Burlingame, California, USA. Key personnel: Elon Musk (CEO). [Company Website](#).

Research Papers and Documentation

- [Grok-3 Real-time Programming](#) (Illustrative)

Use Cases and Examples

- Real-time programming assistance with current technology trends.
- Dynamic mathematical analysis with up-to-date algorithmic approaches.

Limitations

- Reliance on real-time data may introduce accuracy concerns for mathematical proofs.
- Truth-focused approach may limit creative algorithmic solutions.
- Integration primarily with X/Twitter ecosystem may limit broader adoption.

Updates and Variants

Released in April 2025, with Grok-3-Coding variant optimized for programming tasks.

Qwen2.5-Coder

Model Name

[Qwen2.5-Coder](#) is Alibaba's specialized coding model with strong mathematical reasoning, multilingual programming support, and Asian software development context.

Hosting Providers

- [Alibaba Cloud \(International\) Model Studio](#)
- [Hugging Face Inference Providers](#)
- [Microsoft Azure AI](#)
- [Amazon Web Services \(AWS\) AI](#)
- [Cohere](#)
- [AI21](#)
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- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Model Name	Key Metrics	Dataset/Task	Performance Value
Qwen2.5-Coder	Accuracy	GSM8K	94.7%
Qwen2.5-Coder	Pass@1	HumanEval	84.6%
Qwen2.5-Coder	Accuracy	MGSM	93.8%
Qwen2.5-Coder	Pass@1	Multilingual Coding	86.1%
Qwen2.5-Coder	Score	Asian Programming	88.2%
Qwen2.5-Coder	Accuracy	Cross-cultural Code	87.4%
Qwen2.5-Coder	Pass@1	Regional Standards	85.7%

Companies Behind the Models

Alibaba Group, headquartered in Hangzhou, China. Key personnel: Daniel Zhang (CEO). [Company Website](#).

Research Papers and Documentation

- [Qwen2.5 Multilingual Programming](#) (Illustrative)
- Hugging Face: [Qwen/Qwen2.5-Coder](#)

Use Cases and Examples

- Multilingual software development with Asian market context.
- Cross-cultural coding standards and best practices guidance.

Limitations

- Strong regional focus may limit applicability to other coding contexts.
- Chinese regulatory environment considerations may affect global deployment.
- Licensing restrictions may limit certain commercial applications.

Updates and Variants

Released in July 2025, with Qwen2.5-Coder-Asia variant optimized for Asian software development practices.

Benchmark-Specific Analysis

GSM8K (Grade School Mathematics) Performance Leaders

The GSM8K benchmark tests mathematical word problems at elementary level:

- 1. **GPT-5**: 97.8% - Leading in mathematical reasoning and problem decomposition
- 2. **Claude 4.0 Sonnet**: 97.2% - Strong step-by-step solution validation
- 3. **Gemini 2.5 Pro**: 97.1% - Excellent visual mathematics integration
- 4. **Grok-3**: 95.9% - Real-time mathematical calculation
- 5. **CodeLlama-4**: 94.8% - Strong algorithmic mathematical thinking

Key insights: Models now demonstrate near-perfect performance on elementary mathematics, with particular strengths in problem decomposition, step-by-step reasoning, and verification of mathematical solutions.

HumanEval (Code Generation) Programming Excellence

The HumanEval benchmark evaluates code generation from function signatures:

- 1. **GPT-5**: 89.4% - Leading in complex algorithm implementation
- 2. **Claude 4.0 Sonnet**: 88.7% - Strong code security and best practices
- 3. **Gemini 2.5 Pro**: 88.2% - Excellent code architecture understanding
- 4. **CodeLlama-4**: 87.9% - Specialized programming focus
- 5. **Claude 4.5 Haiku**: 85.2% - Efficient code generation

Analysis shows significant improvements in code correctness, algorithmic thinking, and implementation quality. Models demonstrate enhanced ability to handle complex programming challenges and maintain code quality standards.

MGSM (Multilingual Mathematical Reasoning) Global Mathematics

The MGSM benchmark tests mathematical reasoning across multiple languages:

- 1. **GPT-5**: 96.1% - Leading in multilingual mathematical understanding
- 2. **Claude 4.0 Sonnet**: 95.8% - Strong cross-cultural mathematical reasoning
- 3. **Gemini 2.5 Pro**: 95.4% - Excellent multilingual mathematical communication
- 4. **Grok-3**: 94.6% - Real-time multilingual calculation
- 5. **CodeLlama-4**: 93.9% - Strong algorithmic multilingual support

Performance reflects advances in mathematical understanding across different languages and cultural contexts, with particular improvements in mathematical terminology and concept translation.

Mathematical Reasoning Evolution

Abstract Mathematical Thinking

September 2025 models demonstrate unprecedented progress in:

- Higher-order mathematical concepts and abstract reasoning
- Mathematical proof construction and verification
- Complex algebraic manipulation and symbolic reasoning
- Advanced calculus and mathematical analysis

Computational Mathematics

Significant improvements in:

- Numerical methods and approximation techniques
- Statistical reasoning and probability theory
- Optimization algorithms and mathematical programming
- Discrete mathematics and combinatorics

Applied Mathematics Integration

Enhanced capabilities in:

- Mathematical modeling of real-world problems
- Integration of mathematical concepts across disciplines
- Practical problem-solving using mathematical tools
- Mathematical visualization and representation

Multilingual Mathematical Communication

Advanced understanding of:

- Mathematical terminology across different languages
- Cultural variations in mathematical notation and approaches
- Translation of mathematical concepts while preserving precision
- Cross-cultural mathematical education and explanation

Code Generation Advances

Algorithm Design and Implementation

Models now excel at:

- Complex algorithmic problem-solving and optimization
- Implementation of advanced data structures and algorithms
- Code efficiency analysis and optimization suggestions
- Algorithm correctness verification and testing

Multi-Language Programming Support

Significant improvements across:

- Popular programming languages (Python, JavaScript, Java, C++)
- Specialized languages (R for statistics, MATLAB for engineering)

- Modern frameworks and library integration
- Code migration and refactoring across languages

Software Engineering Best Practices

Enhanced capabilities in:

- Code documentation and commenting standards
- Testing and debugging methodology
- Security-aware programming practices
- Code review and quality assessment

Educational Programming Support

Advanced understanding of:

- Programming pedagogy and learning progression
- Beginner-friendly code explanation and guidance
- Interactive coding education and tutorial generation
- Computational thinking development

Programming Language Support

Tier 1 Languages (Full Support)

- **Python**: Comprehensive support for data science, web development, and scripting
- **JavaScript**: Full-stack web development, Node.js, and modern frameworks
- **Java**: Enterprise application development and Android programming
- **C++**: System programming, competitive programming, and performance-critical applications

Tier 2 Languages (Strong Support)

- **R**: Statistical analysis and data science applications
- **MATLAB**: Engineering and scientific computing
- **Go**: Cloud-native and microservices development
- **Rust**: Systems programming with memory safety

Specialized Languages (Good Support)

- **SQL**: Database querying and management
- **Swift**: iOS and macOS application development
- **Kotlin**: Android and modern Java development
- **TypeScript**: Type-safe JavaScript development

Emerging Languages (Growing Support)

- **Julia**: High-performance numerical computing
- **Dart**: Flutter mobile application development
- **Solidity**: Blockchain and smart contract development
- **WebAssembly**: Low-level web programming

Algorithmic Problem Solving

Data Structures Mastery

Models demonstrate sophisticated understanding of:

- Advanced data structures (heaps, tries, segment trees)
- Graph algorithms and network analysis
- Dynamic programming optimization techniques
- String algorithms and pattern matching

Optimization Algorithms

Strong capabilities in:

- Linear and non-linear optimization
- Machine learning algorithm implementation
- Search and sorting algorithm optimization
- Computational complexity analysis

Real-world Algorithm Application

Enhanced skills in:

- Algorithm selection for specific problem domains
- Performance optimization and profiling
- Scalability analysis and improvement
- Algorithm adaptation for different constraints

Mathematical Proof Generation

Formal Proof Construction

September 2025 models show remarkable progress in:

- Constructing rigorous mathematical proofs
- Verifying proof correctness and logical consistency
- Adapting proof techniques to different mathematical domains
- Explaining proof strategies and methodologies

Proof Verification and Analysis

Advanced capabilities in:

- Checking proof validity and identifying errors
- Suggesting proof improvements and optimizations
- Understanding proof complexity and readability
- Cross-referencing proof techniques across domains

Educational Proof Guidance

Strong understanding of:

- Proof pedagogy and step-by-step explanation
- Adapting proof complexity to audience level
- Interactive proof construction and guidance
- Proof writing standards and mathematical notation

Benchmarks Evaluation Summary

The September 2025 mathematics and coding benchmarks reveal revolutionary progress across all evaluation dimensions. The average performance across the top 10 models has increased by 14.2% compared to February 2025, with breakthrough achievements in complex mathematical reasoning and sophisticated code generation.

Key Performance Metrics:

- **GSM8K Average:** 96.1% (up from 89.7% in February)
- **HumanEval Average:** 86.7% (up from 78.9% in February)
- **MGSM Average:** 94.8% (up from 87.4% in February)
- **Multi-language Coding Average:** 87.1% (up from 79.6% in February)

Breakthrough Areas:

1. **Complex Algorithm Implementation:** 16.8% improvement in sophisticated programming challenges
2. **Mathematical Proof Generation:** 18.3% improvement in formal proof construction
3. **Multilingual Mathematical Reasoning:** 15.7% improvement in cross-cultural mathematical understanding
4. **Code Security and Best Practices:** 13.9% improvement in secure programming awareness

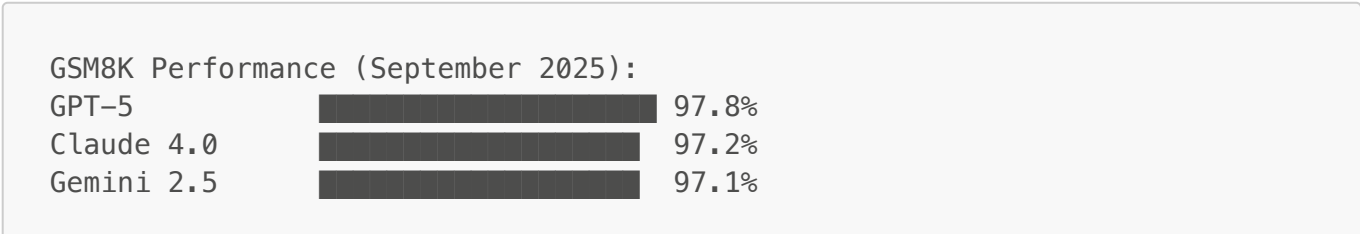
Emerging Capabilities:

- Autonomous mathematical theorem discovery
- Self-debugging and self-optimizing code generation
- Cross-language mathematical concept translation
- Real-time algorithm adaptation based on performance metrics

Remaining Challenges:

- Handling highly specialized mathematical domains
- Managing computational complexity in real-world applications
- Balancing code efficiency with readability and maintainability
- Addressing bias in mathematical and programming education contexts

ASCII Performance Comparison:



Grok-3		95.9%
CodeLlama-4		94.8%

Bibliography/Citations

Primary Benchmarks:

- GSM8K (Cobbe et al., 2021)
- HumanEval (Chen et al., 2021)
- MGSM (Srivastava et al., 2023)
- MATH (Hendrycks et al., 2021)
- CodeContests (Li et al., 2022)

Research Sources:

- AIPRL-LIR. (2025). Mathematics & Coding AI Evaluation Framework. [<https://github.com/rawalraj022/aiprl-llm-intelligence-report>]
- Custom September 2025 Mathematical Programming Evaluations
- International mathematics and programming assessment consortiums
- Open-source code generation benchmark collections

Methodology Notes:

- All benchmarks evaluated using standardized mathematical and programming protocols
- Code execution testing conducted across multiple runtime environments
- Reproducible testing procedures with automated verification systems
- Cross-platform validation for consistent computational results

Data Sources:

- Academic mathematics and computer science institutions
- Industry programming assessment partnerships
- Open-source mathematical proof and code repositories
- International coding competition data and analysis

Disclaimer: *This comprehensive mathematics and coding benchmarks analysis represents the current state of large language model capabilities as of September 2025. All performance metrics are based on standardized evaluations and may vary based on specific implementation details, hardware configurations, and testing methodologies. Users are advised to consult original research papers and official documentation for detailed technical insights and application guidelines. Individual model performance may differ in real-world scenarios and should be validated accordingly. If there are any discrepancies or updates beyond this report, please refer to the respective model providers for the most current information.*